Homework – 2D FDM solver

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The goal of this exercise is to build your own 2D FDM solver for Poisson-type equations on Cartesian grids. The nodal equations, the linear indexing $k = (j-1)\,n_x + i$ and the coefficients k_x, k_y, k_c are those of the FDM 2D slides (equation numbers in this text refer to them).

1 Meshgrid is your friend

- Define two arrays `x` and `y` to store the x- and y-coordinates of your grid nodes. Create and visualize a 5×7 nodes grid with $L_x = L_y = 1$ using the `meshgrid` function:

```
[X,Y] = meshgrid(x,y);
```

- Now you can plot (and check) your grid using

```
plot(X,Y,'b.')
```

  You should get something similar to Fig. 1 (where the node coordinates have been added using the `text` function in a `for` loop).

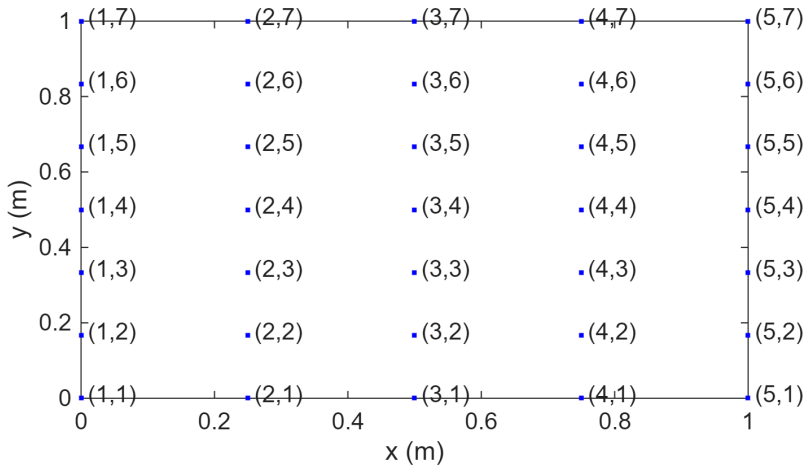


Figure 1: Our first 2D grid.

2 Your first 2D FDM solver

The goal of this step is to create a function that solves Poisson-type problems on a 2D grid, with zero Dirichlet BCs on every side of the domain and a uniform (i.e., constant) source term $t(x, y) = t$. This is the formulation of the problem:

$$\begin{cases} \dfrac{\partial^2\varphi}{\partial x^2} + \dfrac{\partial^2\varphi}{\partial y^2} = t, & \\ \varphi(x,0) = \varphi_S = 0 & \text{South edge,} \\ \varphi(0,y) = \varphi_W = 0 & \text{West edge,} \\ \varphi(L_x,y) = \varphi_E = 0 & \text{East edge,} \\ \varphi(x,L_y) = \varphi_N = 0 & \text{North edge.} \end{cases} \tag{1}$$

Create an m-function `FDM_2D_s1` with the following I/O scheme:

- input: a `geom` structure containing info on the geometry and the discretization (L_x, L_y, number of nodes in the x- and y-directions); a spatially uniform source term `t`;
- output: `phi` (computed values of the unknown function), `X`, `Y` ("meshgridded" nodal coordinates).

Run the code with the following settings:

```
geom.nx = 25;
geom.Lx = 1;

geom.ny = 25;
geom.Ly = 1;
t = -1;

[phi,X,Y] = FDM_2D_s1(geom,t);
```

Use the `surf` function to plot the results:

```
surf(X,Y,phi)
```

You should obtain something like Fig. 2.

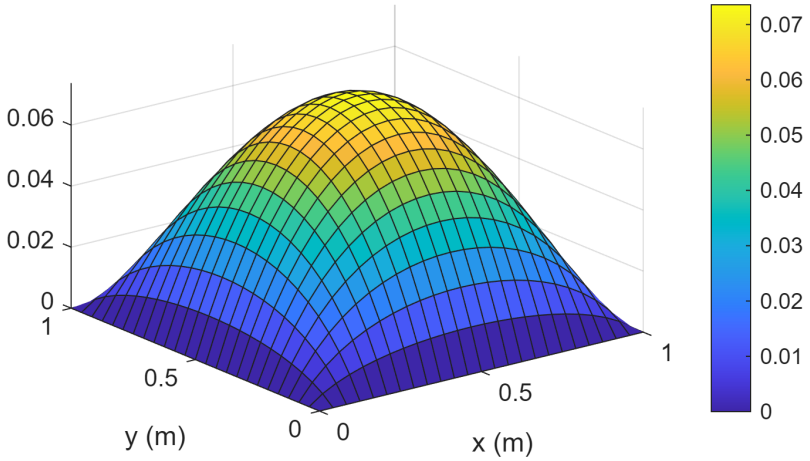


Figure 2: `FDM_2D_s1`: solution with $t = -1$.

Suggestion: you can start from the template `src/04_fdm_2D/FDM_2D_s1.m` in the course repository (lines marked with `TODO` are to be completed; as it is, the function runs and returns placeholder outputs):

```
function [phi,X,Y] = FDM_2D_s1(geom,t)
    % Remember to explain your functions for others, but mainly for yourself in the future!
    %   geom: ...
    %   t: ...
    %   phi = ...
    %   X,Y = ...
    %   example of usage: ...
```

```
    %
    % Lines marked with TODO are to be completed. As it is, the function
    % runs and returns placeholder outputs (phi = 0 everywhere).

    % extract the info you need from geom
    nx = geom.nx; % number of nodes along x
    ny = geom.ny; % number of nodes along y
    % TODO: Lx, Ly, dx, dy

    % use meshgrid to create the X and Y matrices
    X = []; Y = []; % TODO: [X,Y] = meshgrid(...)

    % preallocate [K] and {rhs} for speed using the zeros function
    % TODO: K = ...
    % TODO: rhs = ...

    % pre-define the coefficients of the [K] matrix (see the slides)
    kx = 0; % TODO: horizontal coefficient
    ky = 0; % TODO: vertical coefficient
    kc = 0; % TODO: central coefficient

    % loop for [K] and {rhs} assembly
    for j = 1:ny % number of nodes along the y-direction
        for i = 1:nx % number of nodes along the x-direction
            % linear node index (row by row)
            k = (j-1)*nx + i;

            if i==1 || i==nx || j==1 || j==ny % boundary nodes
                % TODO: K(k,k) = ...
                % TODO: rhs(k) = 0; % boundary condition
            else
                % internal nodes: five-point stencil
                % TODO: K(k,k) = ...
                % TODO: K(k,k+1) = ...
                % TODO: K(k,k-1) = ...
                % TODO: K(k,k+nx) = ...
                % TODO: K(k,k-nx) = ...

                % TODO: rhs(k) = ...
            end
        end
    end

    % solve the linear system using \ (backslash)
    phi = zeros(ny,nx); % TODO: phi = ...

    % the obtained phi array has to be "reshaped" to get an ny x nx matrix,
    % like X and Y: look up the "reshape" function in the MATLAB documentation

end
```

2.1 Verification

Now we verify the output of our code by comparing the numerical solution with an analytical expression.
By applying the method of manufactured solutions, one can show that, for a unit square domain with zero Dirichlet BCs and $t(x, y) = -8\pi^2 \sin(2\pi x) \sin(2\pi y)$, the exact solution is

$$\varphi_{\text{analytical}} = \sin(2\pi x)\sin(2\pi y), \tag{2}$$

which you can define as a function handle:

```
phi_exact = @(x,y) sin(2*pi*x).*sin(2*pi*y);
```

Problem: so far the code accepts only uniform right-hand sides.

- Modify the `FDM_2D_s1` function to accept right-hand sides defined via an anonymous function, e.g.:

  ```
  t = @(x,y) -8*pi^2*sin(2*pi*x).*sin(2*pi*y);
  ```

Check that the numerical solution is consistent with the analytical one, using the following settings:

```
geom.nx = 25;
geom.Lx = 1;

geom.ny = 25;
geom.Ly = 1;
```

By plotting the numerical and the analytical solutions you should get Fig. 3.

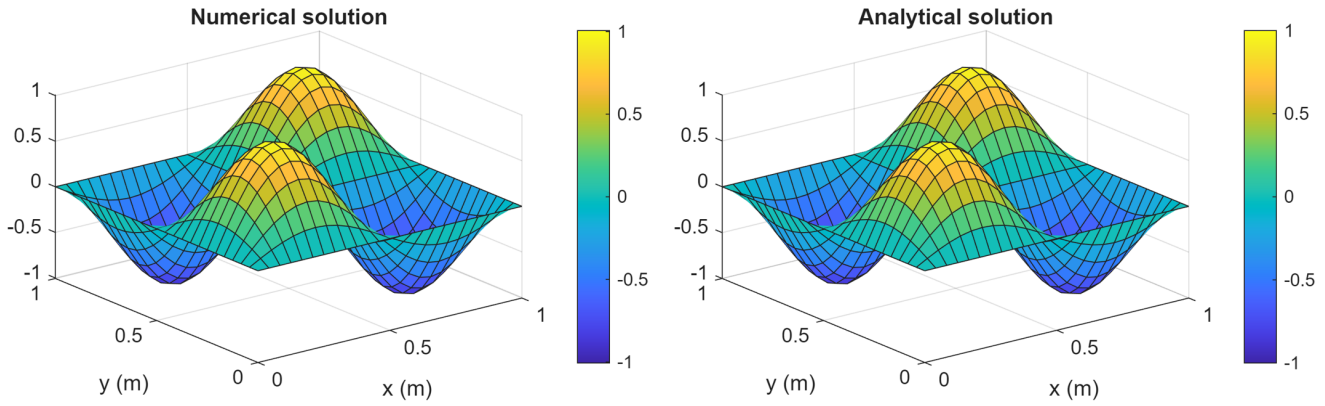


Figure 3: Numerical and analytical solutions for the manufactured source term.

Now we want to check that the accuracy of our code improves when we increase the number of nodes.

- Plot the *error*, i.e. the difference between the numerical and the analytical solutions:

  ```
  err = phi - phi_exact(X,Y);
  ```

- Check that (as expected) the error decreases when you increase `geom.nx` and `geom.ny`. How fast? Compare with the $O(\Delta x^2, \Delta y^2)$ truncation error of the five-point stencil.

2.2 Backward compatibility

Originally, our solver was designed to accept only constant (uniform) values of the source term t. We have now extended it to support spatially varying right-hand sides defined through function handles. However, for backward compatibility, we still want the solver to accept constant values as well.
The solver should handle **both** situations:

- if the user provides a constant value (e.g. `t = 3`), the code should treat it as a uniform source term;
- if the user provides a function handle (e.g. `t = @(x,y) ...`), the solver should evaluate it at the nodal coordinates.

To achieve this flexibility, we can add a small input-normalization step inside `FDM_2D_s1`. Check whether `t` is already a function handle; if not, wrap it into one:

```
% Ensure t is a function handle
if ~isa(t, "function_handle")
    t_value = t;            % copy the constant
    t = @(x,y) t_value;     % wrap it into a constant function
end
```

Check that the trick works by running again the `t = -1` example of Section 2.

3 Improving boundary conditions

Now we want to extend the capabilities of our code by allowing more flexible boundary conditions. Instead of fixing a single BC type for all boundaries, we want the user to specify, independently for each side of the domain, which boundary condition applies and what numerical value is imposed. To do that, store the boundary conditions in a structured variable `BC`, where each field corresponds to one side of the rectangular domain:

- `BC.S` – South boundary
- `BC.W` – West boundary
- `BC.N` – North boundary
- `BC.E` – East boundary

Each of these fields is itself a small structure containing two pieces of information:

- the boundary condition *type* (`'D'` for Dirichlet or `'N'` for Neumann);
- the boundary condition *value*, i.e. the constant value imposed by that BC.

This choice of organization has several advantages:

- It makes the solver easier to extend later: additional BC types (Robin conditions, spatially varying data...) can be added by simply updating the structure, without modifying the overall interface.
- It allows the user to specify different kinds of BCs for each side without changing the numerical code inside the solver.

When assembling the matrix $[K]$ and the vector $\{rhs\}$, the key idea is to recognize that **different nodes require different stencil formulas**, depending on their position in the domain. At every iteration of the (i, j) loop you should:

1. Compute the linear index k associated with node (i, j) (nodes numbered row by row):

$$k = (j-1)\, n_x + i, \qquad \text{neighbours: } (i \pm 1, j) \to k \pm 1, \quad (i, j \pm 1) \to k \pm n_x.$$

2. Determine whether the node is:
   - a **corner** (S–W, S–E, N–W, N–E),
   - an **edge** node (South, West, East, North),
   - or an **interior** node.
3. Apply the appropriate finite-difference stencil. The nodal equations are collected in the slide "FDM 2D – nodal equations" of the FDM 2D slides:
   - interior node: five-point stencil, Eq. (4);
   - Dirichlet edge: known value, Eqs. (6), (7);
   - Neumann edge: four-point stencil with the ghost node eliminated, Eqs. (9), (11);

- corners: two Dirichlet edges → average of the two values, Eq. (5); one Dirichlet and one Neumann edge → Dirichlet value kept; two Neumann edges → three-point stencil, Eq. (14).

A compact way to visualize this logic is:

```
for j = 1:ny
    for i = 1:nx
        k = (j-1)*nx + i; % linear index

        if is_corner(i,j)
            % special stencil for corner
        elseif is_edge(i,j)
            % stencil for edges (Dirichlet or Neumann)
        else
            % interior stencil
        end

    end
end
```

- Create a new function `FDM_2D_s2` and use it to solve the following problem, with Dirichlet and Neumann BCs:

$$\begin{cases} \dfrac{\partial^2 \varphi}{\partial x^2} + \dfrac{\partial^2 \varphi}{\partial y^2} = 1, & \\ \varphi(x,0) = 1 & \text{South edge,} \\ \varphi(0,y) = -1 & \text{West edge,} \\ \left.\dfrac{\partial \varphi}{\partial x}\right|_E = 0 & \text{East edge,} \\ \left.\dfrac{\partial \varphi}{\partial y}\right|_N = 0 & \text{North edge,} \end{cases} \tag{3}$$

with $n_x = n_y = 25$.
According to this formulation, the BC structure should be set to:

```
BC.S.type = 'D';
BC.S.val = 1;

BC.W.type = 'D';
BC.W.val = -1;

BC.E.type = 'N';
BC.E.val = 0;

BC.N.type = 'N';
BC.N.val = 0;
```

Run the code and check your solution, which should look like Fig. 4.

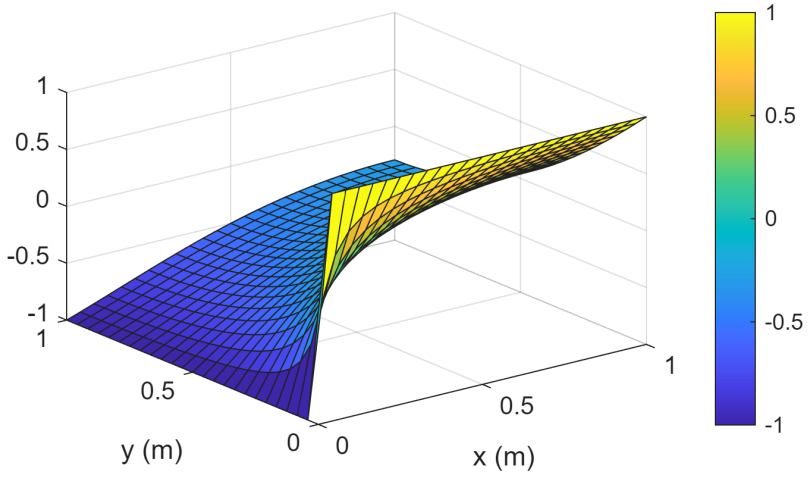


Figure 4: Solution with mixed Dirichlet/Neumann BCs and $t = 1$.